

DevBuddy - Project Requirements Document

1. Project Overview

DevBuddy is an AI-powered developer assistant that allows users to upload documents and interact with an intelligent chat interface. The system consists of a React frontend and a FastAPI backend connected to a PostgreSQL database.

2. Functional Requirements

ID	Requirement
FR-01	Users can upload PDF, DOCX, and TXT files via the chat sidebar.
FR-02	Uploaded files are stored in the backend uploads/ directory.
FR-03	File metadata (name, size, type, date) is persisted in PostgreSQL.
FR-04	Users can view file upload history on the Files page.
FR-05	Users can delete uploaded files from the UI.
FR-06	Users can start new chat conversations.
FR-07	Chat history is maintained per session.
FR-08	The assistant responds to user messages with contextual replies.

3. Non-Functional Requirements

Performance: API response time should be under 500ms for standard requests.

Scalability: The backend architecture should support horizontal scaling.

Security: File uploads must be validated for type and size (max 50MB).

Availability: The system should target 99.9% uptime in production.

4. Tech Stack

Layer	Technology	Version
Frontend	React + Vite + TypeScript	React 19
Backend	FastAPI + Uvicorn	0.115.0
Database	PostgreSQL	16
ORM	SQLAlchemy	2.0.23
Styling	Pure CSS (dark/light)	-